

2157 - Modify Design Parameters

CHANGE load, thickness, height, & width

Load: 300 lbf

Height = 3.00 in

Width = 3.00 in

Thickness

↳ $r = 1.50$ in

Guess: Length should increase drastically because the area is increasing, which is in the numerator of the deflection equation (solved for l), and force is decreasing, which is in the denominator of the same equation.

$$l = \frac{(0.009)(3.1415(1.5)^2)(10.15 \times 10^6)}{300}$$

$$l = 2152.32 \text{ in}$$

CHANGING: Height & Width = 0.100 in

F still 300 lbf

↳ $r = 0.05$ in

Guess: Length should decrease from original design decision

$$l = \frac{0.009 \cdot 3.1415(0.05)^2 \cdot 10.15 \times 10^6}{300}$$

$$l = 2.3915 \text{ inches}$$

Solidworks calculated displacement:

$$2.286e-1 \Rightarrow 0.2286 \times \left(\frac{1 \text{ in}}{25.4 \text{ mm}} \right) = 0.009 \text{ in}$$